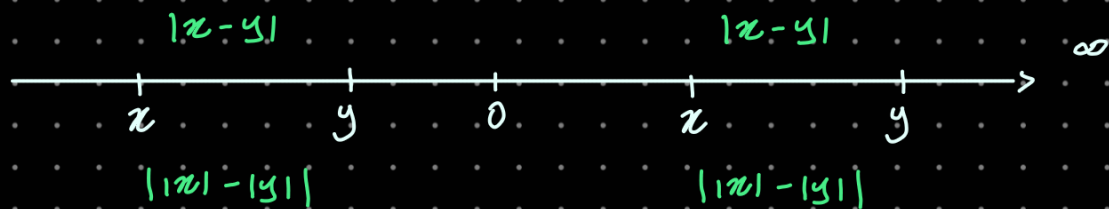


Disug. Triang. Inversa

Se $x, y \in \mathbb{R}$

$$||x| - |y|| \leq |x - y|$$



Proposizione Se $a_n \rightarrow L \in \mathbb{R}$,
allora $|a_n| \rightarrow |L|$

DIM Vogliamo dimostrare:

Sia $\varepsilon > 0$, esiste $N = N(\varepsilon) \in \mathbb{N}$ t.c.

$$||a_n| - |L|| < \varepsilon \quad \forall n \geq N_\varepsilon$$

Sappiamo che per un $\varepsilon > 0$, esiste $M_\varepsilon \in \mathbb{N}$ t.c.

$$|a_n - L| < \varepsilon \quad \forall n \geq M_\varepsilon$$

$$\downarrow$$
$$||a_n| - |L||$$

$$N_\varepsilon := M_\varepsilon$$

Domanda: la proposiz. inversa è vera? NO

ES

$$a_n := (-1)^n$$

$$= \begin{cases} -1 & n \text{ dispari} \\ 1 & n \text{ pari} \end{cases}$$

- $|a_n| = 1 \rightarrow 1$

- $\lim_{n \rightarrow \infty} a_n$ N.E.

- Se $|a_n| \rightarrow 0$ allora

$$a_n \rightarrow 0$$

$\varepsilon > 0$, $\exists N_\varepsilon \in \mathbb{N}$ t.c.

$$|a_n - 0| < \varepsilon \quad \forall n > N_\varepsilon$$

$$|a_n| \nearrow$$

$$||a_n| - 0| \swarrow$$

Decidiamo che:

- Potenze, per $x > 0$

$$f(x) = x^b,$$

???

- le Funz. esponenziali:

$$f(x) := a^x$$

$$a^n \gg n^k$$

$$\lim_{n \rightarrow \infty} \frac{\ln n}{e^n}$$

Dim $n = a^{t_n}$ $t_n \in \mathbb{R}$

$$(t_n := \log_a(n))$$

$$\Rightarrow t_n = \log \sim \sim 1$$

$$\Rightarrow \lfloor t_n \rfloor \xrightarrow{\text{parte intera}} +\infty$$

perché $\lfloor t_n \rfloor \geq t_n - 1 \rightarrow +\infty$

$$\frac{\log_a(n)}{n^\alpha} = \frac{t_n}{(a^\alpha)^{t_n}} \geq \frac{\lfloor t_n \rfloor}{(a^\alpha)^{t_n}}$$

$$n^\alpha = a^{t_n - \alpha} = (a^\alpha)^{t_n}$$

hell naur

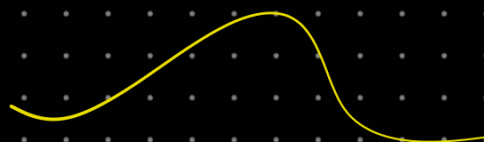
$$\geq \frac{\lfloor t_n \rfloor}{(a^\alpha)^{\lfloor t_n \rfloor - 1}}$$

$\lfloor t_n \rfloor - 1 > t_n$
 $> a^{\alpha t_n}$

$$\lim_{n \rightarrow \infty} \frac{n}{b^n} = 0$$

!

!



Dimostriamo che

$$\lim_{n \rightarrow \infty} \frac{L_{t_n} - 1}{b^{L_{t_n} - 1}} = 0$$

$$K = L_{t_n} - 1$$

Sappiamo che

$$\lim_{k \rightarrow \infty} \frac{k}{b^k} = 0 \quad (b > 1)$$

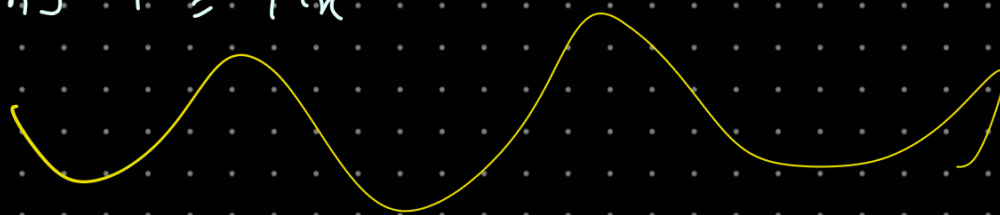
$$\forall m \in \mathbb{N}, \exists M \text{ t.c.}$$

$$\left| \frac{m}{b^m} \right| < \varepsilon \quad \text{se } m \geq M_k$$

$$t_n \rightarrow \infty$$
$$L_{t_n} - 1 \rightarrow +\infty, \exists N \in \mathbb{N}$$

t.c.

$$L_{t_n} - 1 \geq M_k$$



Ricordiamo

$$n! = 1 \cdot 2 \cdot \dots \cdot n$$

$$\bullet \lim_{n \rightarrow +\infty} \frac{a^n}{n!} = 0 \quad a > 1$$

$$\text{Per } n > a \quad (n > 2)$$

$$\frac{a^n}{n!} = \frac{a}{1} \cdot \frac{a}{2} \cdot \dots \cdot \frac{a}{\lfloor a \rfloor} \cdot \frac{a}{\lfloor a \rfloor + 1}$$

$$\leq a^{\lfloor a \rfloor + 1}$$

$$\leq a^{\lfloor a \rfloor + 1}$$

=

$$\bullet \lim \frac{n!}{n^n}$$

$$n^n \gg n! \gg e^n \gg n^2 \gg \ln n$$

Formula di Stirling

$$n! = 1 \cdot 2 \cdot \dots \cdot n \\ = n^n e^{-n} \sqrt{2\pi n} e^{\frac{\theta_n}{12n}}$$

$$\theta_n \in (0, 1)$$

• Scriviamo

$$\theta_n \underset{\text{simile}}{\sim} b_n \quad \text{se} \quad \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1$$

$$n! \overset{\nearrow}{\sim} n e^{-n} \sqrt{2\pi n}$$

$$\lim_{n \rightarrow +\infty} \sqrt[n]{n!} = +\infty$$

$$\sqrt[n]{n!} \sim (n^n e^{-n} \sqrt{2\pi n})^{\frac{1}{n}}$$

$$\underset{+\infty}{\sim} \underbrace{\underbrace{n e^{-1}}_{\substack{\in \mathbb{R} \\ \downarrow \\ 1}} \underbrace{\sqrt[n]{2\pi}}_{\downarrow 1} \underbrace{\sqrt[n]{n}}_{\downarrow 1}}_{\substack{\hookrightarrow e^{-1} \\ +\infty}}$$